2.5 Multiple-Angle and Product-to-Sum Formulas

Name _____ Period ____ Date _____

Pre-Calculus 2.5 · Student sheet

I can…

- use double-angle formulas to rewrite, evaluate, and solve.
- use power-reducing and half-angle formulas.
- use product-to-sum and sum-to-product formulas.

Part 1 · Double or not?

Use a calculator in degree mode. Round to three decimal places.

Expression	Value	Compare with	Value	Equal?
sin 2(30°)		2 sin 30°		
sin 60°		2 sin 30° cos 30°		
cos 60°		$\cos^2 30° - \sin^2 30°$		
cos 60°		$1 - 2\sin^2 30°$		

So sin 2u = _____ and cos 2u = _____

Part 2 · Notes: fill in the blanks

Double-angle: sin 2u = _____ tan 2u = _____

cos 2u = _____ = _____ = _____

Power-reducing: $\sin^2 u$ = _____ $\cos^2 u$ = _____

Half-angle: sin(u/2) = _____ cos(u/2) = _____ tan(u/2) = _____

The ± sign in a half-angle formula depends on the quadrant of _____.

Product-to-sum: sin u cos v = _____

Part 3 · Examples (work them with the class)

Example 1

Find all solutions in $[0, 2\pi)$: $2\cos x + \sin 2x = 0$

Example 2

$\cos u = \frac{5}{13}$ with $\frac{3\pi}{2} < u < 2\pi$. Find sin 2u, cos 2u, and tan 2u.

Example 3

Use a double-angle formula to rewrite $6 \sin x \cos x$.

Example 4

Rewrite $\sin^4 x$ in terms of first powers of cosines of multiple angles.

Example 5

Find the exact value of $\sin 105°$.

Example 6

a. Rewrite $\cos 5x \sin 4x$ as a sum or difference.

b. Rewrite $\sin 5x + \sin 3x$ as a product.

Part 4 · Partner practice

Partner 1

$\sin u = \frac{3}{5}$ with $\frac{\pi}{2} < u < \pi$. Find $\sin 2u$.

Partner 2

Find the exact value of $\cos 22.5°$.